Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Reference point

Meaning in my words How would you explain reference point to a friend?	**How can you tell?** A sign says “5 m east.” What place do we need to identify?
Example or sketch How would you mark a cart 3 m west of a doorway chosen as zero?	**Non-example and why** If I choose the table as zero instead, did the cart actually move?

Known quantity

Meaning in my words What makes a number a known quantity in a problem?	**How can you tell?** A cart starts at rest and is timed for 3 s. Which two values can you write down?
Example or sketch A sensor reads +4 m at the start. How would you enter that known value?	**Non-example and why** A friend lists a guessed final speed among the given values. Where should it go?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Model assumption

Meaning in my words How would you explain model assumption to a friend?	**How can you tell?** What must be true to predict another 2 m each second from a cart's earlier motion?
Example or sketch Write an assumption beside a straight-line forecast for XJ.	**Non-example and why** Is tomorrow's camera reading an assumption once we measure it?

Range

Meaning in my words How would you explain range to a friend?	**How can you tell?** What positions are included in 12 ± 2 m?
Example or sketch How would you mark the forecast on a number line?	**Non-example and why** A partner says only 12 m counts as a match. What do the endpoints show?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.



My marks mean

Two reports. One sighting.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare position reports that use different zeros.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

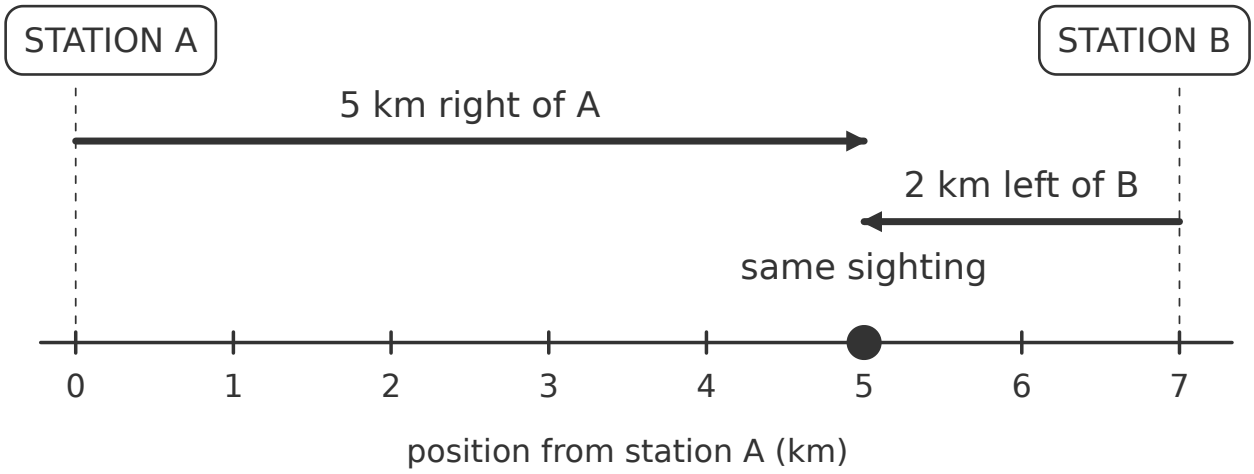
Use one axis for both stations. Give dispatch the position from A, then name what a future forecast still needs.

MAP BROADCASTER ☐

Show both stations and zeros on the locator map. Explain whether the two reports describe different locations.

Point to a clue. Keep your first marks; add the update using the color key.

Two observatories file the same sighting: "**5 km** right of station A" and "**2 km** left of station B." B is **7 km** right of A.



01 OUR FIRST CALL

Do the reports agree? Give one location and explain how the map supports it.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What number changes in the new frame? What physical location stays the same?

Switch speakers. Are the two stations using the same zero?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks